Examiner only

7. *(a)* Potassium has three stable isotopes – ^{39}K, ^{40}K and ^{41}K.

(i) Compare the nuclei of each of these isotopes. [1]

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(ii) Use the information to calculate the relative atomic mass, A_r, of potassium.

Record your answer to **three** significant figures. [3]

Isotope	Relative isotopic mass	% in sample
^{39}K	39	93.1
^{40}K	40	0.0122
^{41}K	41	6.88

$$A_r = \frac{(\text{mass} \times \%\ \text{isotope 1}) + (\text{mass} \times \%\ \text{isotope 2}) + (\text{mass} \times \%\ \text{isotope 3})}{100}$$

A_r = ...

Examiner only

(b) Lithium lies above potassium in the Periodic Table.

(i) Give **two** similarities and **two** differences between the reactions of potassium and lithium with water. [2]

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(ii) Write a balanced symbol equation for the reaction between potassium and water. [3]

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